

0.1 State Variable

C_v is specific heat for constant volume.

p, V, T, n, U . Cannot write Q , because you can pull heat in and out. The relationships of these:

$$\begin{aligned}pV &= nRT \\U &= \frac{3}{2}nRT \\&= N_{\text{molecules}} E_{\text{kinetic}}\end{aligned}$$

We can describe this in a pV diagram. When you see a pV diagram that has an area under the curve, you can calculate it with:

$$W = \int p dv$$

How this is derived:

$$\begin{aligned}W &= \int F dv \\&= Ap \frac{dv}{A} \\&= \int p dv\end{aligned}$$

Now talking about non state variable:

$$U = Q - W \tag{1}$$

0.2 Processes

isobaric $p = \text{constant}$

isochoric $V = \text{constant}$

isothermic $T = \text{constant}$

adiabatic $Q = \text{constant}$

To maintain the same pressure (isobaric), apply heat and increase temperature. To maintain the same volume (isochoric), cool it down. To keep the temperature the same (isothermal), keep it in thermodynamic equilibrium while changing the pressure / volume via a heat bath. To keep the heat the same, insulate the system.

1. Isobaric.

- $W = \int p dV = p \Delta V = nR \Delta T.$
- $Q = nC_p \Delta T$
- $\Delta U = Q - W = n(c_p - R) \Delta T = nc_v \Delta T$

2. Isochoric.

- $W = \int p dV = 0$
- $Q = nc_v \Delta T$
- $\Delta U = Q = nc_v \Delta T = \frac{3}{2} nRT \rightarrow c_v = \frac{3}{2} R$

3. Isothermal.

- $W = \int p dV = \int_{v_i}^{v_f} (nRT) dV = nRT \ln(\frac{V_f}{V_i})$
- $\Delta U = 0 \rightarrow Q - W = 0 \rightarrow Q = W$

4. Adiabatic

- $\Delta Q = 0 \rightarrow \Delta U = -\Delta W \rightarrow du = -dW$
- Just look in lecture notes for derivation.
- $pV^\gamma = pV^{\frac{c_p}{c_v}}$ constant